Page 2: Introduction – The Link Between Electricity and Magnetism

For a long time, electricity and magnetism were thought to be separate phenomena. However, in 1820, a Danish scientist named **Hans Christian Oersted** made a groundbreaking discovery that changed everything.

Oersted's Experiment (The Key Discovery)

- **Setup:** He placed a compass needle (a small magnet) near a current-carrying wire.
- **Observation:** When the current was switched on, the compass needle deflected. When the current was switched off, the needle returned to its original position. When the direction of current was reversed, the needle deflected in the opposite direction.
- **Conclusion:** This experiment proved for the first time that **an electric current produces a magnetic field around it**.

This discovery established the fundamental link between electricity and magnetism, giving rise to the branch of physics called **electromagnetism**.

Key Concept:

- A current-carrying conductor acts like a magnet.
- The magnetic field is produced only when current flows. No current = no magnetic field.
- The direction of the magnetic field depends on the direction of the current.

Page 3: Magnetic Field and Field Lines

What is a Magnetic Field?
A magnetic field is the region surrounding a magnet (or a current-carrying conductor) in which its magnetic force can be experienced (felt) by another magnet or magnetic material.

Magnetic Field Lines
To visualize this invisible field, we use the concept of magnetic field lines. They are imaginary lines that represent the path a north pole would take if placed in the field.

Properties of Magnetic Field Lines (Very Important for Exams):

1. **Direction:** They emerge from the **North Pole** and merge at the **South Pole** outside the magnet. Inside the magnet, they travel from South to North, thus forming **closed loops**.

2. **No Intersection:** Two magnetic field lines never intersect each other. If they did, it would mean two directions of the magnetic field at the same point, which is impossible.
3. **Strength and Closeness:** The strength of the magnetic field is proportional to the closeness of the field lines.
    - Where lines are **crowded**, the field is **strong**.
    - Where lines are **spread apart**, the field is **weak**.
4. **Tangents:** The tangent drawn to a field line at any point gives the direction of the magnetic field at that point.

Page 4: Magnetic Field Due to a Straight Current-Carrying Conductor

When current passes through a straight wire, it creates a magnetic field around it.

Pattern of Field Lines:
The magnetic field lines are **concentric circles** centered on the wire. They lie in a plane perpendicular to the wire.

- **Plane of circles:** Perpendicular to the wire.
- **Center of circles:** The wire itself.

Strength of the Field (B)
The strength of the magnetic field (B) produced depends on:

1. **Current (I):** Higher the current, stronger the field. **($B \propto I$)**
2. **Distance (r):** Farther from the wire, weaker the field. **($B \propto 1/r$)**

Direction of the Field – Right-Hand Thumb Rule
To find the direction of the magnetic field lines, we use a simple rule.

Maxwell's Right-Hand Thumb Rule:

- Imagine you are holding the current-carrying wire in your **right hand**.
- Point your **thumb** in the direction of the **current**.
- The direction in which your **fingers curl** around the wire gives the direction of the **magnetic field lines**.

Example: If current flows upwards, the magnetic field lines are anticlockwise as seen from above. If current flows downwards, the field lines are clockwise.

Page 5: Magnetic Field Due to a Circular Loop (Coil)

If we bend a straight wire into a circular loop and pass current through it, the magnetic field produced has a specific pattern.

Pattern of Field Lines:

- The magnetic field lines are **concentrated near the center** of the loop.
- At the center, the field lines are **straight and perpendicular** to the plane of the loop.
- The field lines appear to emerge from one face of the coil and merge into the other face.

Concept of Magnetic Polarity of a Loop:
A current-carrying loop behaves like a thin disc magnet with two faces:

- The face from which the field lines **emerge** behaves as the **North Pole**.
- The face into which the field lines **enter** behaves as the **South Pole**.

Factors Affecting the Magnetic Field at the Center:
The strength of the magnetic field at the center of a circular coil depends on:

1. **Current (I):** Directly proportional.
2. **Radius (r):** Inversely proportional (smaller radius gives stronger field).
3. **Number of Turns (n): Directly proportional.** This is a key concept. If you have a coil with multiple turns (n), the magnetic field produced is **n times** the field due to a single loop. This is because the current in each circular turn contributes to the field.

Page 6: Magnetic Field Due to a Solenoid

A **solenoid** is a long coil consisting of a large number of close-packed insulated turns of wire wrapped in the shape of a cylinder.

The Magnetic Field Pattern:

- The field lines inside the solenoid are **parallel, straight, and close together**. This indicates that the magnetic field is **uniform** (same strength and direction) at all points inside the solenoid.
- The field outside the solenoid is weak and similar to that of a bar magnet.

The Solenoid as an Electromagnet:
When current is passed through a solenoid, it behaves exactly like a **bar magnet**.

- One end of the solenoid acts as the **North Pole** (from which field lines emerge).

- The other end acts as the **South Pole** (into which field lines enter).

How to Determine the Polarity?
Use the **Clock Face Rule** (also known as the Ampere's Swimming Rule):

- Look at one end of the solenoid.
- If the current flows in the **clockwise** direction, that end is the **South Pole**.
- If the current flows in the **anti-clockwise** direction, that end is the **North Pole**.

Advantages of a Solenoid over a Permanent Magnet:

1. The strength of the magnetic field can be increased or decreased by changing the current.
2. The polarity (North-South) can be reversed by reversing the direction of current.
3. The magnet exists only when current flows; it can be switched off.

Page 7: Force on a Current-Carrying Conductor in a Magnetic Field

We know that a magnet exerts a force on another magnet. Oersted showed that a current-carrying wire (which acts like a magnet) also exerts a force on a compass needle. Therefore, logically, a magnet should also exert a force on a current-carrying wire.

The Observation:
When a current-carrying conductor is placed in an external magnetic field (like between the poles of a horseshoe magnet), it experiences a mechanical force.

Cause of the Force:
This happens due to the interaction between two magnetic fields:

1. The magnetic field produced by the current-carrying conductor itself.
2. The external magnetic field in which it is placed.
   These two fields interact, resulting in a net force on the conductor.

Factors Affecting the Force:
The force (F) is maximum when the conductor is perpendicular to the magnetic field and zero when it is parallel. It depends on:

- Strength of the magnetic field (B)
- Current flowing through the conductor (I)
- Length of the conductor (L)

Direction of Force – Fleming's Left-Hand Rule
To find the direction of force, we use Fleming's Left-Hand Rule.

Fleming's Left-Hand Rule:

- Stretch the **thumb**, **forefinger**, and **central finger** of your **left hand** so that they are mutually perpendicular to each other (like an X, Y, and Z axis).
- The **Forefinger** points in the direction of the **Field** (magnetic field, from North to South).
- The **Central finger** points in the direction of **Current** (conventional current, from positive to negative).
- The **Thumb** then points in the direction of **Force** or motion.

This rule is fundamental to understanding how electric motors work.

Page 8: Electromagnetic Induction – The Reverse Effect

Oersted showed that electricity can produce magnetism. About a decade later, **Michael Faraday** discovered the reverse effect: **magnetism can produce electricity**.

Faraday's Experiment:
Faraday connected a coil of wire to a galvanometer (a device to detect small currents). He then placed a bar magnet near the coil.

- **Observation 1:** When the magnet was **moved towards** the coil, the galvanometer showed a deflection, indicating a current in the circuit.
- **Observation 2:** When the magnet was **held stationary** inside the coil, the galvanometer showed **no deflection**.
- **Observation 3:** When the magnet was **moved away** from the coil, the galvanometer again showed a deflection, but in the **opposite direction**.

Conclusion:

- A current is induced in a coil whenever there is a **change in the magnetic field lines** (magnetic flux) linked with the coil.
- This phenomenon is called **Electromagnetic Induction**.
- The induced current exists only as long as the magnetic field is changing (relative motion between magnet and coil).

The induced current is produced not by the presence of a magnetic field, but by the *change* in the magnetic field.

Page 9: Direction of Induced Current and Its Application

Fleming's Right-Hand Rule (Generator Rule)
Just as we have a rule to find the direction of force (motor), we have a rule to find the direction of induced current (generator).

Fleming's Right-Hand Rule:

- Stretch the **thumb**, **forefinger**, and **central finger** of your **right hand** so that they are mutually perpendicular.
- The **Thumb** points in the direction of **Motion** of the conductor relative to the field.
- The **Forefinger** points in the direction of the **Field** (magnetic field, North to South).
- The **Central finger** then points in the direction of the **Induced Current**.

Applications of Electromagnetic Induction:

1. **Electric Generator:** Converts mechanical energy into electrical energy. It uses the principle that when a coil is rotated in a magnetic field, the magnetic flux linked with it changes, inducing a current.
2. **Transformers:** Used to change the voltage of alternating current (will be discussed next).

Page 10: Important Devices and Summary

1. Electric Motor (Converts Electrical → Mechanical Energy)

- **Principle:** A current-carrying conductor placed in a magnetic field experiences a force (Fleming's Left-Hand Rule).
- **Working:** In a motor, a coil is placed between magnets. When current flows through the coil, a force acts on its two arms in opposite directions (due to opposite current directions), causing the coil to rotate. A device called a **split-ring commutator** reverses the current direction every half-turn to keep the coil rotating in the same direction.

2. Electric Generator (Converts Mechanical → Electrical Energy)

- **Principle:** Electromagnetic Induction (Fleming's Right-Hand Rule).
- **Working:** A coil is rotated mechanically in a magnetic field. As the coil rotates, the magnetic flux linked with it changes, inducing a current in the coil. The direction of induced current changes every half-turn, producing **Alternating Current (AC)** . A

device with slip rings can be used to produce AC, while a commutator can be used to produce Direct Current (DC).

3. AC (Alternating Current) vs. DC (Direct Current)

- **DC:** Current flows in one direction only. (e.g., from a cell or battery).
- **AC:** Current reverses its direction periodically. (e.g., from our home power supply). In India, the frequency of AC is **50 Hz**, meaning it changes direction 50 times per second.

4. Domestic Electric Circuits

- In our homes, circuits are connected in **parallel**. This ensures that each appliance gets the same voltage (220V) and can be operated independently.
- **Live Wire (Red/Brown):** Carries current from the supply.
- **Neutral Wire (Black/Blue):** Completes the circuit.
- **Earth Wire (Green):** A safety wire that connects the metal body of an appliance to the ground, providing a low-resistance path for current in case of a short circuit, preventing electric shock.